

Are there other methods of evaluating the determinant of a 3×3 matrix?

IB Mathematics Analysis and Approaches Internal Assessment

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Introduction and Rationale

Why is linear algebra important? In today's modern society, linear algebra is integral in solving many problems. On the surface level, it is the study of vectors, vector spaces, linear transformations, and systems of linear equations. These ideas help serve to innovate a multitude of technological advancements and support tackling complex real-world challenges.

In particular, one of its primary applications lies in the manipulation and analysis of data to have efficient numerical computations—useful in various fields. Moreover, our society's reliance on machine learning and data science further emphasizes the importance of linear algebra in our contemporary society. Through the utilization of linear algebraic operations, there are many techniques to enable us to extract meaningful data. However, outside of data analysis and machine learning, linear algebra finds itself useful in other academic fields. Civil Engineers rely on the optimization of complex systems to efficiently optimize bridges or skyscrapers.

The essence of linear algebra allows individuals and communities to comprehend the increasingly technologically advanced and data-driven world we live in. While not everyone may study linear algebra, it remains undisputed that it is an indispensable discipline that serves to spark innovation and progress in the 21st century.

For the last semester of my senior year, outside of IB Mathematics, I have been studying linear algebra at my local community college—Los Angeles City College. I'm particularly fascinated with linear operations and the wide ways of the discipline being applied. From a simple sudoku game to analyzing data from the United States Pentagon, it proves to be a universal area of knowledge that is increasingly becoming more in demand. One of the concepts that I've both had difficulty learning but have also enjoyed understanding is evaluating determinants. Linear algebra is especially important because unlike single variables or even multivariable calculus, linear algebra stands on its own, evaluating dimensions that one cannot even imagine to fathom.

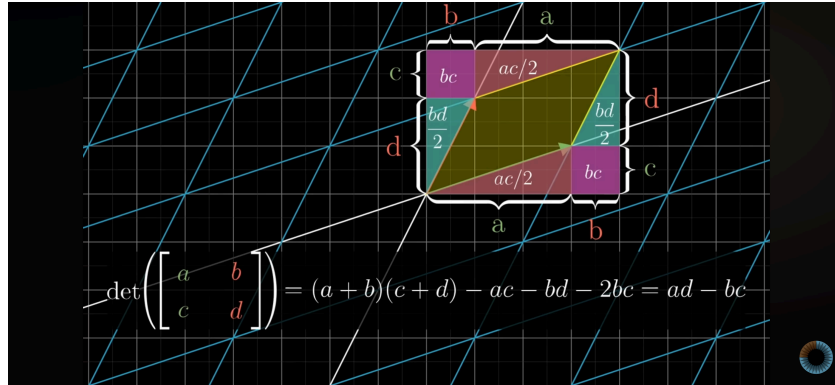


Figure #1: 3Blue1Brown Youtube Video: The determinant

For example, Figure #1, highlights both the formulaic and visual representations of evaluating a determinant of a 2x2 matrix. Thus, finding the area of the parallelogram in 2D. Thus, I ask the inquiry—can you visualize evaluating the determinant of a 10-dimensional space? More likely than not, you are unable to do so. However, the power of linear algebra allows us to manipulate data mathematically despite not being able to even fathom visualizing what your doing looks like.

For this exploration, I will be attempting to find other ways of evaluating the determination of a simple 3x3 matrix. By doing so, I hope to discover an efficient method of finding the **determinant** that visually makes sense to me.

Aim and Methodology

$$\begin{matrix} & \begin{matrix} 1 & 2 & \dots & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{matrix} & \left[\begin{array}{cccc} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{array} \right] \end{matrix}$$

Figure #1: Equation for matrices.

This exploration will primarily utilize matrices in a 3x3 size. The size of a matrix is often notated through the symbols $m \times n$ with m representing the row number and n representing the column number in a matrix. A matrix can be both a rectangular or square matrix with an array of numbers, symbols, or even expressions placed onto the rows and columns. Each value is located somewhere on the matrix. For example, a value located in the second row and first column would be denoted as a (2,1) with the first 2 being the row placement and the second 1 being the column placement.

$$\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

Figure #2: A symbolic version of 3x3 Matrix.

In this investigation, I aim to utilize the 3x3 Matrix in Figure #2 to comprehend the different methods of finding determinants through various international methods and possibly develop my method of solving determinants. I plan to only use these specific notations throughout my mathematical analysis—A1, A2, A3, B1, B2, B3, C1, C2, and C3.

The three theorems I will be utilizing are each separate—American, Indian, and Russian.

$$A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Figure #3: American Theorem of finding the determinant of a 3x3 matrix.

Figure #3, is a popular theorem that many people studying—especially in United States institutions—utilize. However, other theorems allow you to find the determinant of a 3x3 matrix.

Figure #4: Rule of Sarrus of finding the determinant of a 3x3 matrix.

Figure #4, is a theorem developed by French mathematician Pierre Frédéric Sarrus which is a unique method to find the determinant of a 3x3 matrix by diagonally finding the sum of products.

$$\begin{aligned}
 &+ a_{1,1}a_{2,2}a_{3,3} \\
 &+ a_{1,2}a_{2,3}a_{3,1} \\
 &+ a_{1,3}a_{2,1}a_{3,2}
 \end{aligned}
 \qquad
 \begin{aligned}
 &- a_{1,3}a_{2,2}a_{3,1} \\
 &- a_{1,2}a_{2,1}a_{3,3} \\
 &- a_{1,1}a_{2,3}a_{3,2}
 \end{aligned}$$

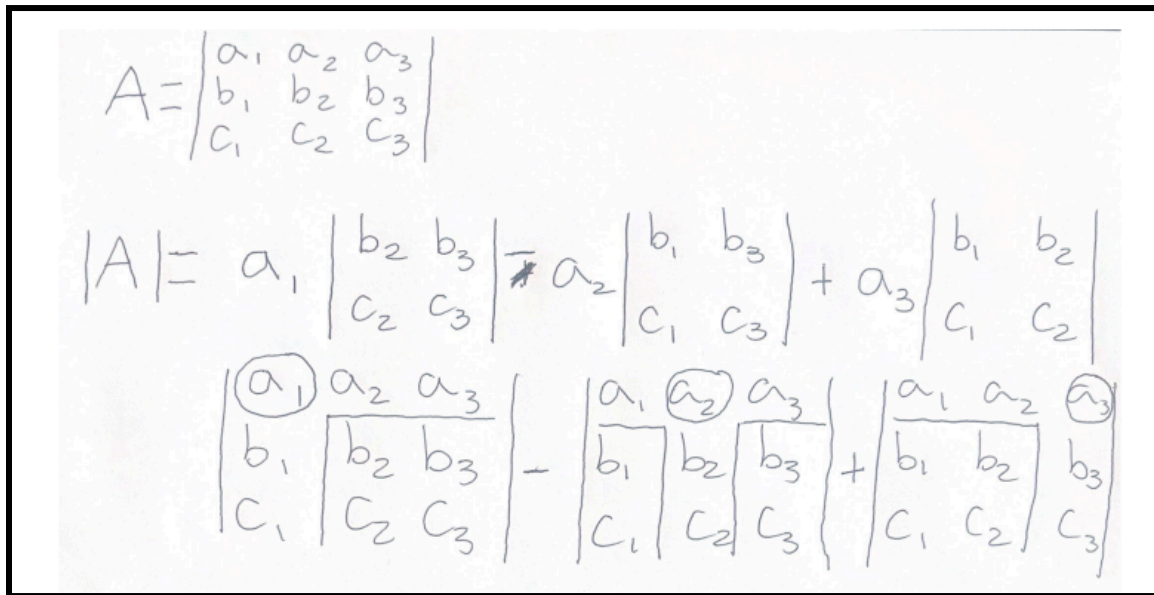
Figure #5: Russian method of finding a 3x3 determinant.

Figure #5, is a Russian theorem that when researching online seemed like a very uncommon and unused method but is the theorem that my East Russian Linear Algebra taught in my class.

Ultimately I plan to manipulate and analyze these theorems to see the pattern and fundamental understanding to hopefully develop and create my method of finding the determinant.

Mathematical exploration

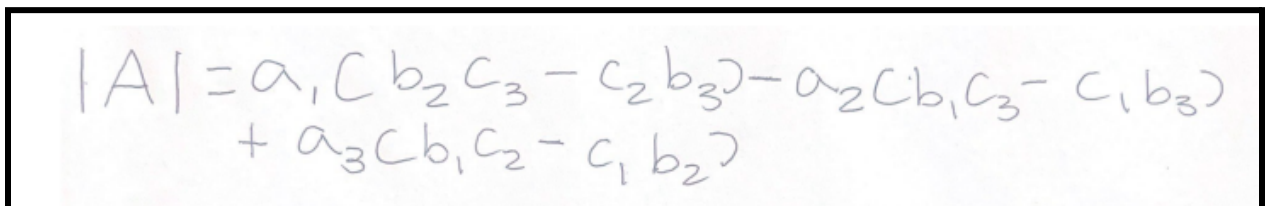
Firstly, I will be using the most common computational formula to evaluate 3x3 matrices determinants which is in in Figure #3 through variable notations provided in Figure #2.



The image shows a handwritten derivation of the determinant of a 3x3 matrix A . At the top, the matrix is defined as $A = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$. Below this, the determinant is expanded using the American Theorem: $|A| = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$. The final line shows the same expansion with the first elements of each minor circled: $\begin{vmatrix} \textcircled{a_1} & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} - \begin{vmatrix} a_1 & \textcircled{a_2} & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & a_2 & \textcircled{a_3} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

Figure #6: The American Theorem to evaluate the determinant of matrices.

It is important to recognize that 3 minors must be evaluated. Minors are smaller matrices within the overall original matrix. In Figure #6, we see that 3 minors can be evaluated with the formula in Figure #1—AD-BC.



The image shows a handwritten formula for the determinant of a 3x3 matrix with the minors evaluated: $|A| = a_1(b_2c_3 - c_2b_3) - a_2(b_1c_3 - c_1b_3) + a_3(b_1c_2 - c_1b_2)$.

Figure #7: Determinate with the minors evaluated.

Ultimately, this is the most standard equation you can get through using the American Theorem. Now I will be evaluating the Rule of Sarrus as another theorem to find the fully evaluated formula.

$$\begin{array}{ccc|cc}
 a_1 & a_2 & a_3 & a_1 & a_2 \\
 b_1 & b_2 & b_3 & b_1 & b_2 \\
 c_1 & c_2 & c_3 & c_1 & c_2
 \end{array}$$

$$a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2$$

$$-c_1 b_2 a_3 - c_2 b_3 a_1 - c_3 b_1 a_2$$

Figure #8: Rule of Sarrus to evaluate the determinate.

This theorem multiplies the left and middle column of the matrix and places it onto the right of the matrix. By doing this the theorem's next step is to do the sum of the products of each diagonal line.

$$|A| = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - c_1 b_2 a_3 - c_2 b_3 a_1 - c_3 b_1 a_2$$

Figure #9: Fully simplified equation of finding the determinant using the Rule of Sarrus.

Hence, these two theorems have allowed evaluating the determinate with general variable notations that I have created. Thus, based on inductive reasoning the equations in Figure #9 and Figure #7 should be equal.

$$\begin{array}{ll}
 \text{American Theorem} & \text{Sarrus Theorem} \\
 a_1(b_2c_3 - c_2b_3) - a_2(b_1c_3 - c_1b_3) & a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 \\
 + a_3(b_1c_2 - c_1b_2) & = -c_1b_2a_3 - c_2b_3a_1 - c_3b_1a_2
 \end{array}$$

Figure #10: Placing the equations equal to each other

Now I have proven that these two theorems—taught in different regions of the world—should be equal. However, I will be doing distributive property on the formula in the American Theorem to fully see how equal these two formulas are.

American Theorem

$$\begin{aligned}
 & a_1(b_2c_3 - c_2b_3) - a_2(b_1c_3 - c_1b_3) \\
 & + a_3(b_1c_2 - c_1b_2) \\
 \hline
 & a_1b_2c_3 - a_1c_2b_3 - a_2b_1c_3 + a_2c_1b_3 \\
 & + a_3b_1c_2 - a_3c_1b_2 \\
 \hline
 & a_1b_2c_3 + a_2c_1b_3 + a_3b_1c_2 \\
 & - a_1c_2b_3 - a_2b_1c_3 - a_3c_1b_2
 \end{aligned}$$

Figure #11: Equation simplified through distributive property.

Distributive Property Formula

$a(b+c)$ $= ab + ac$	$a(b-c)$ $= ab - ac$
$-a(b+c)$ $= -ab - ac$	$-a(b-c)$ $= -ab + ac$

Figure #12: Distributive property formula

In Figure #11, I simplified the equation to demonstrate its equivalence to the equation on the right-hand side in Figure #10. This simplification was achieved by applying the distributive property technique as highlighted in Figure #12. Thus, by utilizing both the American and Sauss theorems, I have proven that both techniques for evaluating the determinant are equivalent, despite involving different steps.

Now, I will be evaluating the determinate of a 3x3 matrix by using the Russian method to highlight how this technique, although different, provides the same equation.

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} - \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3 - c_1b_2a_3 - b_1a_2c_3 - a_1b_3c_3$$

Figure #13: Russian Method of evaluating the determinant.

Now, by using three different methods of finding the determinant of a 3x3 matrix, I have showcased how these three methods often provide the equivalent equation necessary for one to find the determinant of the matrix.

$$a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3 - c_1b_2a_3 - b_1a_2c_3 - a_1b_3c_3$$

Figure #14: General formula to find the determinant of the 3x3 matrix.

By utilizing all three methods, I have determined that although the steps to find the determinant may be different, the generic formula that allows each method to find the determinate arises from the formula highlighted in Figure #14. It is important to note that the steps to do so may differ and depending on the method of manipulating the 3x3 matrix, the final formula may look slightly different but must be equivalent to the general formula written in Figure #14.

Furthermore, I will now be developing my method of finding the determinant. Currently, my favorite method is the Russian method that my Linear Algebra has taught me but there are some facets of the steps through using that method which makes the evaluation of the determinant a bit tedious. Thus, I will make my own method in which I can find the determinant of a 3x3 matrix. Hence, I will base the foundation of my method of finding the determinant of the matrix on utilizing the Russian theorem but I will put my twist to it.

Step 1: $a_1 b_2 c_3 - c_1 b_2 a_3$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Step 2: Switch each value in the corner of the matrix diagonally. RE: $c_1 \leftrightarrow a_3$ $a_1 \leftrightarrow c_3$.

$$\begin{array}{cc} \text{Original} & \text{Transformed} \\ \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} & \rightarrow \begin{vmatrix} c_3 & a_2 & c_1 \\ b_1 & b_2 & b_3 \\ a_3 & c_2 & a_1 \end{vmatrix} \end{array}$$

Step 3: Draw triangles in each corner of the matrix with the new transformed matrix.

$$\begin{vmatrix} c_3 & a_2 & c_1 \\ b_1 & b_2 & b_3 \\ a_3 & c_2 & a_1 \end{vmatrix}$$

Step 4: Find the product of each triangle and organize the triangles in the negative diagonal as negative and the positive diagonal as positive.

$$= \begin{vmatrix} c_3 & a_2 & c_1 \\ b_1 & b_2 & b_3 \\ a_3 & c_2 & a_1 \end{vmatrix} \begin{vmatrix} c_1 & a_2 & b_3 \\ b_1 & c_2 & a_3 \end{vmatrix} + \begin{vmatrix} c_3 & a_2 & c_1 \\ b_1 & b_2 & b_3 \\ a_3 & c_2 & a_1 \end{vmatrix} \begin{vmatrix} b_1 & a_2 & c_3 \\ a_1 & c_2 & b_3 \end{vmatrix}$$

Figure #15: Steps of solving a determinant through utilizing my own method,

I similarly utilized the concepts of triangles that the Russian method used. However, while the Russian method in Figure #5, draws triangles across the diagonal, I made my triangles both equilateral but also rest on the diagonal while the triangles drawn through the Russian method are often perpendicular to the diagonal.

Step 5: Combine the values in step 2 and 5.

$$a_1b_2c_3 - c_1b_2a_3 + c_1a_2b_3 + b_1c_2a_3 - b_1a_2c_3 - a_1c_2b_3$$

step #1 Step #4

Figure #16: Final step to finding the determinant of a 3x3 matrix.

When looking at the equation written in Figure #16, it is important to note that the final equation is equivalent to the general formula in Figure #14.

Now that I have my method of finding the determinant of a 3x3 matrix, I need to provide ethos to my method by checking the method by utilizing the properties of determinants law—Determinant of the Transpose—to see if my method provides the accurate determinant.

Theorem 3.2.6: Determinant of the Transpose

Let A be a matrix where A^T is the transpose of A . Then,

$$\det(A^T) = \det(A)$$

Figure #17: Properties of determinants—Determinant of the Transpose

The property of the determinant I will be utilizing is the Determinant of the Transpose method. This proof states that the determinant of a matrix must have the same determinant if it were to be transposed.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2 \times 3} \quad A^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}_{3 \times 2}$$

Figure #18: Matrix A and the transposed version of Matrix A.

The transpose of a matrix is found by interchanging its rows into columns or columns into rows. Thus, one of the properties of determinants states that the determinant of the original matrix and the determinant of the respective transposed matrix must have the same determinant. Thus, I will be using my method of finding the determinant and checking if the determinant of the original matrix equals the determinant of the transposed matrix. I will be using a random 3x3 matrix with real numbers in it.

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

Figure #19: 3x3 Matrix.

Firstly, I will be finding the transposed version of this matrix which is in Figure #20.

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 4 & 2 \\ 2 & 1 & 3 \end{pmatrix}$$

Figure #20: Transposed version of the 3x3 matrix in Figure #19.

Now, I will be finding the determinant of both Figure #19 and Figure #20—checking if they give the same values. Ultimately I will be using the proof that $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ which is in Figure #17.

Handwritten calculation of the determinant of matrix $A = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 2 & 3 \end{vmatrix}$.

Step #1: Sarrus' rule calculation.

$$\begin{vmatrix} 2 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow 2 \cdot 4 \cdot 3 - 1 \cdot 4 \cdot 2 = 24 - 8 = \underline{16}$$

Step #2: Cofactor expansion along the first row.

$$\begin{vmatrix} 2 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 3 & 1 & 1 \\ 1 & 4 & 1 \\ 2 & 2 & 2 \end{vmatrix}$$

Step #3: Cofactor expansion along the first row of the 3x3 minor.

$$\begin{vmatrix} 3 & 1 & 1 \\ 1 & 4 & 1 \\ 2 & 2 & 2 \end{vmatrix} - \begin{vmatrix} 3 & 1 & 1 \\ 1 & 4 & 1 \\ 2 & 2 & 2 \end{vmatrix}$$

Step #4: Calculation of the 2x2 minors.

$$1 \cdot 1 \cdot 1 + 2 \cdot 2 \cdot 1 - 3 \cdot 1 \cdot 1 - 2 \cdot 2 \cdot 1 = 1 + 4 - 3 - 4 = \underline{-2}$$

Step #5: Final calculation.

$$\underline{16} - \underline{2} = 14 \quad \text{step \#5}$$

$$\det(A) = 14$$

Figure #21: The determinant of Figure #19.

In Figure #21, I utilized the steps of my method in Figure #15 to find the determinant and ultimately got 14 as the answer. Now, I will check using the Transpose property to see if the determinant would be the same if Matrix A were transposed as highlighted in the matrix in Figure #20.

$$A = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 4 & 1 \\ 1 & 2 & 3 \end{vmatrix} \quad A^T = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 4 & 2 \\ 2 & 1 & 3 \end{vmatrix}$$

Step #1

$$A^T = \begin{vmatrix} \cancel{2} & \cancel{1} & \cancel{1} \\ 1 & 4 & 2 \\ 2 & 1 & 3 \end{vmatrix} \quad 2 \cdot 4 \cdot 3 - 2 \cdot 4 \cdot 1$$

$$\quad \quad \quad \downarrow \quad \quad \downarrow$$

$$\quad \quad \quad 24 - 8 = \underline{16}$$

Step #2

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & 4 & 2 \\ 2 & 1 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 3 & 1 & 2 \\ 1 & 4 & 2 \\ 1 & 1 & 2 \end{vmatrix}$$

Step #3

$$\begin{vmatrix} 3 & 1 & 2 \\ 1 & 4 & 2 \\ 1 & 1 & 2 \end{vmatrix} - \begin{vmatrix} 3 & 1 & 2 \\ 1 & 4 & 2 \\ 1 & 1 & 2 \end{vmatrix}$$

Step #4

$$1 \cdot 1 \cdot 1 + 1 \cdot 2 \cdot 2 - 3 \cdot 1 \cdot 1 - 2 \cdot 2 \cdot 1$$

$$1 + 4 - 3 - 4 = \underline{-2}$$

Step #5

$$16 - 2 = 14$$

$$\det(A^T) = 14$$

Figure #22: The determinant of Figure #20.

Through utilizing the transposed version of the matrix, I was able to find the determinant to be 14. This value is the same as the determinant of the matrix when the matrix wasn't transposed.

Ultimately, through utilizing my new method of finding the determinant that I have created, I was able to find the determinant of Matrix A in Figure #19. I've also proved that this method is foolproof by highlighting how it passes the Determinant of a Transpose test.

Conclusion and Evaluation

Throughout this exploration, I've discovered various applications and methods for proving and evaluating the determinant of a 3x3 matrix. As I've delved into the nature and formulaic processes of finding this determinant, I've realized a limitation: its definitive conclusion. Regardless of the method used, whether Russian, American, Indian, or one I created, all approaches ultimately lead to the same general form:

$$a_{11}a_{22}a_{33}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}-a_{13}a_{22}a_{31}.$$

Yes, while you can manipulate the processes of finding the determinant in your own way, whether it be multiplying in your preferences, categorizing the linear system of equations through multiple minor matrices, or simply organizing the positive and negative signs differently. No matter how it's done, there is a formulaic method of finding these determinants of the 3x3 matrix. Thus, more likely than not, while there could be many different approaches to finding the determinant of a 3x3 matrix, it simply all comes down to what the mathematician prefers. However, that is the beauty of math. While one method may seem arbitrarily harder for others, it may be easier for some. Thus, having many different approaches and methods to do one task is useful as it provides mathematicians with more clarity and understanding of the problem in front of them.

However, it is important to note that in a society where technology can perform all the computations of a matrix, this entire exploration was conducted entirely by hand. While computers have the capability to solve determinants using just one method and can do so faster than any human, this exploration symbolizes my creative pursuit of understanding the properties of matrices and determinants, rather than merely finding a method to solve determinants. Hence, it is crucial for future researchers, mathematicians, and students to comprehend the conceptual ideas of their fields of discipline. Although computers can likely find the determinant of a 100x100 matrix faster than any human, the significance of mathematics lies not just in solving equations quickly, but in our ability as humans to interpret the data provided by machines and guide society toward a better path by optimizing the utilization of the collected data.

Thus, for the future of this exploration to be heightened, I would like to look beyond the methodology and formulas to solve systems of equations or matrices and to discover ways to apply this knowledge for the betterment of society.

$$a_{31}(a_{12}a_{23} - a_{22}a_{13}) + (a_{32}(a_{21}a_{13} - a_{23}a_{11}) + a_{33}(a_{11}a_{22} - a_{21}a_{12}))$$

$$\begin{vmatrix} a & a & a \\ a & a & a \\ \textcircled{a} & a & a \end{vmatrix} + \begin{vmatrix} a & a & a \\ a & a & a \\ a & \textcircled{a} & a \end{vmatrix} + \begin{vmatrix} a & a & a \\ a & a & a \\ a & a & \textcircled{a} \end{vmatrix}$$

$$(a_{12}a_{23}a_{31}) - (a_{31}a_{22}a_{13}) + (a_{21}a_{32}a_{13}) - (a_{32}a_{23}a_{11}) + (a_{11}a_{22}a_{33} - a_{21}a_{12}a_{33})$$

Manipulating Russian Method

$$(a_{11} \cdot a_{22} \cdot a_{33}) + (a_{21} \cdot a_{32} \cdot a_{13}) + (a_{12} \cdot a_{23} \cdot a_{31}) - (a_{31} \cdot a_{22} \cdot a_{13}) - (a_{32} \cdot a_{23} \cdot a_{11}) - (a_{21} \cdot a_{12} \cdot a_{33})$$

$$\cancel{a_{33}(a_{22}a_{11} - a_{21}a_{12})}$$

$$a_{22}(a_{11}a_{33} - a_{31}a_{13}) + a_{21}(a_{32}a_{13} - a_{12}a_{33}) + a_{23}(a_{12}a_{31} - a_{32}a_{11})$$

$$\begin{vmatrix} a & a & a \\ a & \textcircled{a} & a \\ a & a & a \end{vmatrix} + \begin{vmatrix} a & a & a \\ \textcircled{a} & a & a \\ a & a & a \end{vmatrix} + \begin{vmatrix} a & a & a \\ a & a & a \\ a & a & \textcircled{a} \end{vmatrix}$$

Figure #23: Personal notes

Ultimately, this exploration allowed me to discover my own method of evaluating the determinate of a 3x3 matrix. Although it may not be practical to some people, I was fascinated and enjoyed indulging in the different methods of manipulating the general form of evaluating the determinant. In Figure #23, we can see my attempts to manipulate the data, find common factors, and try to see any patterns in general. This exploration allowed me to the interconnectedness of the various methodologies from different areas of the world to

ultimately answer one problem—finding the determinant of a 3x3 matrix. Moreso, I was eventually able to create my own method as highlighted in the steps in Figures #15 and #16.

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